

Combinatorial Algorithms (Math 482), 2026 Spring Problem Set 12

1. **Online hitting set for intervals.** We are given the set $S = \{1, 2, \dots, n\}$, and an adversary reveals a sequence of intervals $C = (A_1, A_2, \dots, A_m)$. We maintain a hitting set $H \subset S$: Initially, $H = \emptyset$, when set A_i is revealed, we can add a new point to H ; and we need to ensure that $H \cap A_i \neq \emptyset$ for all intervals revealed so far. The goal is to minimize the size of H .

- (a) Sophia suggests the following online algorithm: When interval A_i is revealed and $H \cap A_i = \emptyset$, add the rightmost point of $S \cap A_i$ to H . Show that the competitive ratio of this algorithm is $\Omega(n)$ by designing an *adversary strategy*: A sequence of intervals for which Sophia's algorithm achieves a ratio $ALG/OPT \geq \Omega(n)$.
- (b) Tammy suggests another online algorithm: When interval A_i is revealed and $H \cap A_i = \emptyset$, add the “middle” point of $S \cap A_i$ to H . The *middle point* is defined as $\lceil (\min(S \cap A_i) + \max(S \cap A_i))/2 \rceil$. Show that the competitive ratio of this algorithm is $\Omega(n)$ by showing that for any sequence of interval, Tammy's algorithm achieves a ratio $ALG/OPT \leq O(\log n)$.

2. **Online unit covering.** In the *offline* unit covering problem, we are given n real numbers $x_1, \dots, x_n$, and we need to find the minimum number of intervals of unit length (of the form $[a, a + 1]$) such that each of $x_1, \dots, x_n$ is covered by some interval. In the *online* version of the problem, the real numbers $x_1, \dots, x_n$ arrive one-by-one. We need to maintain a collection $\mathcal{C}$ of unit intervals that jointly cover all points that have arrived so far. We can add new unit intervals to $\mathcal{C}$, but we cannot remove any of them. In step i , when x_i arrives, either x_i is covered by an interval in $\mathcal{C}$, or else we need to add a new interval to $\mathcal{C}$ to cover x_i . Show that the competitive ratio of following algorithm is ∞ .

LEFTMOST: If x_i is not covered by any interval in $\mathcal{C}$, add interval $[x_i, x_i + 1]$ to $\mathcal{C}$.

Show that the following two algorithms are 2-competitive.

CENTERED: If x_i is not covered by any interval in $\mathcal{C}$, add $[x_i - \frac{1}{2}, x_i + \frac{1}{2}]$ to $\mathcal{C}$.

GRID: If x_i is not covered by any interval in $\mathcal{C}$, add interval $[\lfloor x_i \rfloor, \lfloor x_i \rfloor + 1]$ to $\mathcal{C}$.

3. **Online unit clustering.** A sequence of real number $x_1, x_2, \dots, x_n$ arrive one-by-one. We need to group these numbers into the minimum number of *clusters* such that each cluster *can be* covered by a unit interval. In step i , when x_i arrives, we either add x_i to an existing cluster (if this cluster with one additional element can still be covered by a unit interval), or create a new cluster that contains x_i only.

Design a 2-competitive online algorithm for this problem. (Describe the algorithm and prove that its competitive ratio is at most 2.)